

Observe that $\mathbf{I} : \mathcal{L}^2 \rightarrow \mathcal{L}^2$ is a monotone operator as soon as h is a nondecreasing map, since

$$\begin{aligned}\langle \mathbf{I}(u) - \mathbf{I}(v), u - v \rangle &= \frac{1}{k(T)} \mathbb{E} \left[(h(Z_T^u) - h(Z_T^v)) \int_0^T G(T-t)(u_t - v_t) dt \right] \\ &= \frac{1}{k(T)} \mathbb{E} \left[(h(Z_T^u) - h(Z_T^v)) (Z_T^u - Z_T^v) \right] \geq 0, \quad u, v \in \mathcal{L}^2.\end{aligned}$$

Then, applying Fubini's Theorem we get

$$\langle \mathbf{III}(u) - \mathbf{III}(v), u - v \rangle = -\mathbb{E} \left[\int_0^T (h'(Z_t^u) - h'(Z_t^v))(Z_t^u - Z_t^v) \frac{dg_t}{k(t)} \right], \quad u, v \in \mathcal{L}^2,$$

so that the monotonicity of the operator $\mathbf{III}$ may not true in general and depends on the dynamics of the endogenous signal g . The particular case where $dg_t = g'(t)dt$ with $g' \geq 0$ (i.e., g is a nondecreasing input curve) and h' is nonincreasing (i.e., the impact function h is concave on the real line) would yield the monotonicity of $\mathbf{III}$, but such assumptions may be too restrictive.

Furthermore, verifying the monotonicity property of the operators $\mathbf{II}$ and $\mathbf{IV}$ in general is not obvious and depends on the form of the kernel G and its first-argument derivative $\partial_x G$.

Case of one exponential for the impact decay. For this reason, in what follows we assume (i) h to be nondecreasing and we restrict our attention to the exponential Volterra kernel, i.e.,

$$G(t, s) = \mathbb{1}_{\{t \geq s\}} a e^{-b(t-s)}, \quad t, s \in [0, T], \quad a > 0, b \geq 0,$$

to study the monotonicity property of the operator $\mathbf{V} := \mathbf{II} + \mathbf{IV}$. In this way, we conveniently use the relation $\partial_x G(t, s) = -bG(t, s)$, for $0 \leq s < t \leq T$, and the fact that $k \equiv a$. Specifically, by Fubini's Theorem and straightforward calculus, we get for any $u, v \in \mathcal{L}^2$

$$\begin{aligned}\langle \mathbf{V}(u) - \mathbf{V}(v), u - v \rangle &= \frac{b}{a} \langle h(Z^u) - h(Z^v), Z^u - Z^v \rangle \\ &\quad + \frac{b}{a} \langle Z^u h'(Z^u) - Z^v h'(Z^v), Z^u - Z^v \rangle \\ &\quad - \frac{b}{a} \mathbb{E} \left[\int_0^T g_t (h'(Z_t^u) - h'(Z_t^v)) (Z_t^u - Z_t^v) dt \right].\end{aligned}$$

Concluding on the nonnegativity of the above quantity in general depends on the endogenous signal g as was the case for $\mathbf{III}$. But if we additionally assume that (ii) $g \equiv 0$ and (iii) $x \mapsto xh'(x)$ is nondecreasing, then clearly $\mathbf{V}$ is monotone, which proves Proposition 2.8.

Case of a sum of exponential time scales for the impact decay. More generally, one may wonder whether we could derive in a similar way sufficient conditions to get the monotonicity property of the operator $\mathbf{V} : \mathcal{L}^2 \rightarrow \mathcal{L}^2$ in the case of a Volterra kernel given by a finite sum of exponentials, i.e.,

$$G(t, s) = \mathbb{1}_{\{t \geq s\}} \sum_{i=1}^n a_i e^{-b_i(t-s)}, \quad t, s \in [0, T], \quad a_i > 0, b_i \geq 0, \quad i \in \{1, \dots, n\}, \quad n \in \mathbb{N}.$$

The answer turns out to be negative. Indeed, assume without loss of generality that the mean-reversion speeds $(b_i)_i$ differ from one another. Then, in this case, $k \equiv \sum_{i=1}^N a_i =: A$ and

$$\partial_x G(t, s) = - \sum_{i=1}^n a_i b_i \exp\{-b_i(t-s)\}, \quad 0 \leq s < t \leq T.$$

For any $i = 1, \dots, n$, we also define the Volterra kernels $G_i(t, s) = \mathbb{1}_{\{t \geq s\}} a_i e^{-b_i(t-s)}$, so that $G(t, s) = \sum_{i=1}^n G_i(t, s)$, $s, t \in [0, T]$, and $\partial_x G(t, s) = - \sum_{i=1}^n b_i G_i(t, s)$, $0 < s < t < T$. Considering the initial input curve $g \equiv 0$, we can write

$$Z^u = \sum_{i=1}^n \mathbf{G}_i u =: \sum_{i=1}^n Z^{i,u}, \quad u \in \mathcal{L}^2.$$